

Abinash Prasana Selvanathan

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Profile

MSc Artificial Intelligence student at Dublin Business School, graduating September 2026, with over a year at Deloitte supporting HIPAA-compliant healthcare IT systems in production. During the MSc I have built end-to-end AI and data systems covering NLP, RAG pipelines, transformer language modelling, machine learning and PostgreSQL analytics. Available full-time from September 2026 for graduate roles and internships across AI, data, software engineering and enterprise technology in Ireland.

Education

Dublin Business School <i>MSc, Artificial Intelligence</i>	Sep 2025 – Expected Sep 2026 <i>Dublin, Ireland</i>
Sri Krishna Adithya College of Arts & Science <i>B.Sc., Computer Technology — First Class Honours, GPA: 7.42/10</i>	2020 – 2023 <i>Coimbatore, India</i>
P.A. Polytechnic College <i>Diploma, Computer Engineering — 69%</i>	2016 – 2019 <i>Pollachi, India</i>

Work Experience

Deloitte <i>Associate Analyst</i>	Nov 2023 – Dec 2024 <i>Hyderabad, India</i>
- Supported HIPAA-compliant healthcare IT systems in production including Epic EMR, ServiceNow and Active Directory, diagnosing and resolving software, access and infrastructure issues while maintaining service levels for clinical users. - Performed root-cause analysis on incident data, identified recurring failure patterns and documented resolutions to support compliance audits and reduce repeat incidents. - Worked with infrastructure and compliance teams on patch management cycles, Active Directory provisioning and application testing across regulated production environments.	

Projects

- RiskRadar – IT Incident SLA Breach Predictor** (GitHub)
 - Built a Python automation pipeline to predict SLA breach risk from IT incident logs. Random Forest classifier achieving 92% accuracy, AUC-ROC 0.9684 and F1 0.887, with 1,569 breach cases correctly flagged. Interactive Streamlit dashboard for triage review and pattern analysis.
 - Tech:** Python · Scikit-learn · Random Forest · Pandas · Feature Engineering · Streamlit
- Dáil LLM – Irish Parliamentary Transformer** (GitHub)
 - Built a character-level transformer from scratch in PyTorch on 4.4 million Dáil Éireann parliamentary speeches (Harvard Dataverse, 1919–2013). Full training pipeline with tokenisation, batching and held-out evaluation: Perplexity 4.07, BLEU 0.0104, 3.2M parameters across 4 layers, 8 heads and 256 dimensions. Four-tab Streamlit dashboard for inference and attention visualisation.
 - Tech:** Python · PyTorch · Transformer Architecture · NLP · Streamlit · Checkpoint Management
- Payments Analytics – PostgreSQL Data Pipeline & Dashboard** (GitHub)
 - Designed a six-table PostgreSQL schema for synthetic commercial transaction data and built the full ingestion pipeline using psycopg2 COPY. Wrote eight SQL queries covering window functions, CTEs, multi-table joins and cohort analysis. Delivered a four-tab Streamlit dashboard covering transaction trends, merchant performance, fraud risk and cohort retention.
 - Tech:** Python · PostgreSQL · psycopg2 · Pandas · Streamlit · Plotly · SQL Analytics
- Clinical RAG – Medical Q&A on MIMIC-IV** (GitHub)
 - Built a RAG pipeline on 331,793 de-identified clinical discharge notes (145,914 patients) accessed via CITI ethics training and PhysioNet Data Use Agreement. Embedded 2,000 documents into FAISS, integrated Flan-T5 for grounded answer generation. Achieved 70% accuracy at 10.91s CPU latency. System declines to answer when relevant context is absent, prioritising correctness over coverage.
 - Tech:** Python · Hugging Face Transformers · FAISS · Clinical NLP · Grounded Inference · RAG Pipeline

Skills

- AI & ML:** PyTorch, Hugging Face Transformers, Scikit-learn, FAISS, Ollama, RAG pipelines, transformer architecture, LSTM/GRU/TCN, NLP, TF-IDF, SHAP, LIME, NLTK
- Data & Analytics:** PostgreSQL, psycopg2, SQL (window functions, CTEs, cohort analysis), Pandas, NumPy, EDA, feature engineering, model evaluation (AUC-ROC, F1, RMSE), XGBoost
- Development:** Python, Flask, Streamlit, Plotly, REST APIs, SQLite, Git, Jupyter, Matplotlib
- Enterprise & Compliance:** ServiceNow, Epic EMR, Active Directory, Microsoft 365, Citrix, HIPAA governance

Certifications & Recognition

- AWS Cloud Practitioner Essentials – Amazon Web Services
- IBM Data Science Professional Certificate – Coursera
- Claude Code in Action – Anthropic
- Software Engineering Job Simulation – JPMorganChase via Forage